INCH-POUND

MIL-B-44433(GL)
5 December 1991

MILITARY SPECIFICATION

BAG, BARRIER, MULTI-LAYER FILM

This specification is approved for use by Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers barrier bags used for preserving chemical protective garments.

1.2 Classification. The barrier bags shall be one of the following types as specified (see 6.2).

Type 1 - Natural finish
Type 2 - Matte finish

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be address to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8105

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SPECIFICATIONS

FEDERAL

A-A-203 - Paper, Kraft, Untreated
PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

FEDERAL

FED-STD-101 - Test Procedures for Packaging Materials
FED-STD-595 - Colors Used in Government Procurement

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 882 - Tensile Properties of Thin Plastic Sheeting
D 1922 - Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method
D 3951 - Standard Practice for Commercial Packaging
F 88 - Seal Strength of Flexible Barrier Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1919 Race Street, Philadelphia, PA 19103-1187.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2.1 Barrier film. The barrier film for the bags shall be a colorless, flexible, unsupported multi-layer film (see 6.4). The film shall conform to the physical requirements specified in table I, when tested in accordance with 4.4.2.

TABLE I. Barrier film Physical requirements

Characteristic	Requirement
Thickness, inch, minimum	0.0045
Water vapor transmission rate (WVTR) g/100 sq in/24 hr, maximum	0.40
Tensile strength, psi, minimum	
Machine	3100
Cross machine	3100
Elongation, percent, minimum	
Machine	250
Cross machine	300
Tear strength, grams, minimum	
Machine	250
Cross machine	400
Puncture, pounds, minimum	9.8

3.2.2 Film finish. The film for type I bags shall have a natural gloss finish. The film for type II bags shall have an opaque matte finish.

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3.3 Bag design. The bags, constructed from barrier film, shall be formed by heat sealing three sides. All heat seal seams shall have a minimum strength of a 8 pounds per linear inch when tested as specified in 4.4.4. Bags shall have a tear, nick, or V-notch, 1/8 inch deep in at least one edge, 2 to 3 inches from the open end of the bag. The notch shall not reduce the heat seal width. The legend "TEAR HERE" with an arrow pointing to the notch shall be printed on the bag with lettering a minimum of 1/8 inch in height.

3.4 Dimensions. Bag dimensions shall be as specified in the end item specification or the contract (see 6.2). The first dimension cited shall be the width measured from the inside edges of the side seams and parallel to the opening. The second dimension cited shall be the length measured from the inside edge of the bottom seam to the opening of the bag. Dimensions shall be specified to the nearest 1/4 inch with a tolerance of $\pm$1/8 inch.

3.5 Label panel. One face of the bag shall have a label panel covering 2/3 of the area. The color of the panel shall be light green approximating color 34094 or tan approximating color 33446 of FED-STD-595.

3.6 Workmanship. The bags shall be uniformly made, free from holes, tears, cuts, creases, wrinkles, or other surface imperfections which might impair their usefulness.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products of supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects listed in 4.4.2 and shall be tested for the characteristics specified in 4.4.3.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this document or applicable purchase document.

4.4.1.1 Material testing. The barrier material shall be tested for the characteristics listed in table II. The lot size shall be expressed in linear yards. The sample unit shall be 3 linear yards full width. The inspection level shall be S-1. Any test failure shall be cause for rejection of the lot. All test reports shall contain the individual values utilized in expressing the final results.

TABLE II. Barrier material testing

Characteristic	Requirement paragraph	Test method	No. of determinations	Results reported as
Thickness, inch	3.2.1	ASTM D 882	5	0.0001 inch
Water vapor transmission rate	3.2.1	FED-STD-101, 2017 (MOCON method)	5	0.001 g/ 100 sq in/ 24 hrs
Tensile Strength	3.2.1	ASTM D 882	5 each direction	10 psi

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TABLE II. Barrier material testing (cont'd)

Characteristic	Requirement paragraph	Test method	No. of determinations	Results reported as
Elongation	3.2.1	ASTM D 882	5 each direction	10 percent
Tear Strength	3.2.1	ASTM D 1922	5 each direction	10 grams
Puncture resistance	3.2.1	FED-STD-101, 2065	5 each orientation	0.1 lb

4.4.2 End item visual examination. The bags shall be examined for the defects listed in table III. The lot size shall be expressed in bags. The sample unit shall be one bag. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects.

TABLE III . End item visual defects

Examine	Defect	Classification Major	Minor
Material	Any cut, tear, hole, or split	101	
	Any delamination of the film	102	
	Creases, sharp wrinkles, or surface imperfections which impair use of bag	103	
	Material not of specified type	104	
Heat seal	Not heat sealed as specified	105	
	Not heat sealed on three sides	106	
	Open seal or seal not continuous	107	
Tear notch	Notch missing or not as specified		201
	Notch reduces heat seal width		202
	Notch not labeled		203
Label panel	Panel missing or not as specified	108	
	Panel not specified color		204
	Size of panel not as specified		205
Dimensions	Any dimension not as specified		206
Workmanship	Not uniformly constructed		207

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4.4.3 End item seam strength testing. Bags shall be tested for seam strength in accordance with ASTM F 88.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping containers fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred unit, shall be 2.5.

Examine	Defect
Preservation	Preservation not as specified Mixed types in same unit pack
Marking	Omitted; incorrect, illegible; of improper size, location, sequence or method of application
Materials	Any component missing, damaged or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping,loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than specified

4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Bags of one type shall be stacked flat. Bags that have a flat unfolded area in excess of 2500 square inches may be folded once, provided dunnage material is placed at the fold to prevent damage.

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5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Five hundred bags, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC, grade V2s of PPP-B-636. Each box shall have the contents covered on the top and bottom with a sheet of 30 pound kraft paper conforming to A-A-203. Each box shall be closed, waterproofed and reinforced in accordance with the appendix of PPP-B-636. Boxes shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Five hundred bags, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each box shall have the contents covered on the top and bottom with a sheet of 30 pound kraft paper conforming to A-A-203. Each box shall be closed in accordance with Method II, as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard shipping containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with the appendix of PPP-B-636.

5.2.3 Commercial packing. Bags of one size only shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), bags of one size only, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallet shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract of purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The barrier bag is intended for preserving chemical protective garments.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and data of this specification
b. Type required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3 and 6.3).
e. Width and length dimensions of bag (see 3.4).
f. Levels of packing (see 5.2).
g. Type and class of unit load required (see 5.2.1).
h. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
i. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Film construction. The following constructions were determined to meet the requirements cited for the barrier material:

a. 0.002 inch Nylon/0.0002 inch polyvinylidene dichloride (PVDC)/0.001 inch low density polyethylene (LDPE)/0.002inch linear low density polyethylene (LLDPE)
b. 0.002 inch Nylon/PVDC/0.0015 inch Copolyer/0.0015 inch Surlyn

Other film constructions will be acceptable if the requirements of this specification are met.

6.5 Subject term (key word) listing.

AUIB
Chemical protective garment
Overpack

Preparing activity:

Army - GL

(Project 8105-A349)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-44433(GL)	1991 December 5

3. DOCUMENT TITLE

BAG, BARRIER, MULTI-LAYER FILM

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)

7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
U.S. Army Natick RD&E Center	(1) Commercial: 508-651-4532 (2) AUTOVON/DSN: 256-4532
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IRT Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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